NSS IIT ROORKEE PRESENTS • SOCIOTHON 2026

SMART AND SUSTAINABLE WASTE MANAGEMENT

Problem Statement & Participant Guide

Date		Team Size
24 October 2026	APJ - 103, 104	1–3 members

1. Background and Problem Description

Waste management is a major environmental and social challenge faced by educational institutions, residential communities, public spaces, and urban and rural areas. Increasing amounts of plastic, paper, food waste, packaging materials, and other waste put pressure on existing collection and disposal systems.

In many places, waste is not segregated properly at the source. Biodegradable waste, recyclable materials, and non-recyclable waste are often mixed together, making recycling difficult and increasing the amount sent to landfills. Inefficient collection systems, limited public awareness, lack of real-time data, and inadequate monitoring further worsen the problem.

A significant portion of waste that could be recycled, composted, or converted into useful resources is discarded. This contributes to pollution, unnecessary resource consumption, higher waste-management costs, and potential health risks. Traditional systems may rely on manual monitoring and fixed collection schedules, which may not respond well to changing waste-generation patterns.

The challenge is to use innovation, technology, and practical problem-solving to make waste management more efficient, affordable, environmentally sustainable, and scalable.

2. Main Problem Statement

How can we design and develop an innovative, affordable, and scalable technology-driven solution that improves waste segregation, collection, recycling, and resource recovery in campuses and local communities?

The proposed solution should address one or more major gaps in the existing waste-management process, be practical to implement in real-world conditions, and demonstrate measurable environmental or social benefits.

3. Areas Teams Can Explore

Teams will work on the same central problem statement but may choose different approaches based on their ideas, skills, and interests. The following are suggested directions, not separate problem statements.

A. AI-Based Waste Identification and Segregation

Develop a system that identifies different types of waste and helps separate them into suitable categories, such as biodegradable, recyclable, and non-recyclable waste.

- An AI model that classifies waste from images.
- A smart bin that identifies waste and guides users to the correct compartment.
- A computer-vision system that detects incorrectly segregated waste.
- A mobile app that tells users which bin to use.

Example technologies: Computer vision, CNNs, YOLO, image classification, IoT sensors.

B. Smart Waste Collection and Route Optimization

Design a system that helps collection teams determine when bins need emptying and which routes are most efficient.

- A dashboard showing bin fill levels.
- Optimized collection routes to reduce travel distance and fuel consumption.
- A system that predicts areas likely to generate more waste.
- A platform for reporting missed collections or overflowing bins.

Example technologies: Route optimization, GIS/maps, IoT, predictive analytics, operations research.

C. Recycling and Waste-to-Resource Solutions

Explore ways to convert waste into useful products or recover valuable materials instead of sending everything to landfills.

- An application connecting waste generators with recyclers.
- A system for managing food waste and compost production.
- A platform connecting scrap collectors, households, and recycling businesses.
- Tracking recyclable material from collection to processing.

Example technologies: Digital marketplaces, material tracking, data analytics, process monitoring.

D. Data-Driven Monitoring of Waste Generation

Build a digital solution that helps institutions and communities understand how much waste they generate, what types they produce, and where improvements are needed.

- A dashboard showing daily or weekly waste-generation trends.
- Comparisons of waste generation across campus buildings or hostels.
- A model forecasting future waste generation.
- Tracking segregation rates, recycling rates, and waste reduction.

Example technologies: Python, machine learning, data visualization, databases, web development.

4. What Makes a Good Solution?

Criterion	What it means
Practicality	Can it be used by a campus, community, municipal body, or waste-management organization?
Affordability	Can it be implemented without excessive investment?
Scalability	Can it expand from one location to multiple buildings or communities?
Sustainability	Does it reduce waste, improve recycling, conserve resources, or lower environmental impact?
Innovation	Does it solve an existing problem in a meaningfully better way?
Measurable impact	Can the team demonstrate or estimate improvements using relevant metrics?

For example, a smart-bin system should explain not only how it detects waste, but also its expected cost, how it handles classification errors, and how it improves the existing process.

5. Expected Deliverables

1. Working Prototype

A functional web or mobile application, AI/ML model, dashboard, simulation, hardware prototype, or another suitable demonstration.

2. Solution Explanation

The problem addressed, proposed approach, target users, and how the solution works.

3. Impact Assessment

Relevant metrics such as waste diverted from landfills, segregation accuracy, collection distance saved, or material recycled.

4. Implementation Plan

Required resources, estimated cost, deployment process, potential challenges, and scaling plan.

5. Final Presentation and Demonstration

A short presentation or live demo explaining innovation, technical approach, feasibility, and expected impact.

These are recommended deliverables for the organizers to adopt; they are not additional official requirements unless included in the final event guidelines.

6. Illustrative Project Ideas

Idea 1 — WasteWise AI

A mobile or web application where users upload a photo of an item and receive its waste category, recommended disposal method, and recycling instructions.

Idea 2 — SmartBin Network

A system that monitors bin fill levels and recommends which bins should be collected first, helping reduce overflowing bins and unnecessary collection trips.

Idea 3 — Waste-to-Value Platform

A platform connecting hostels, canteens, households, recyclers, composting facilities, and scrap collectors to help recover useful materials from waste.

Idea 4 — Campus Waste Intelligence Dashboard

A dashboard tracking waste generation by location, forecasting collection needs, and identifying areas where segregation and recycling need improvement.

These are illustrative ideas, not separate problem statements. Teams should be free to propose different approaches within the same theme.

7. Potential Social and Environmental Impact

- Educational institutions: cleaner campuses, hostels, canteens, and public areas.
- Local communities: improved waste segregation and less unmanaged waste.
- Waste collectors: more efficient collection and better access to recyclable materials.
- Recycling businesses: better-segregated and more traceable materials.
- Local authorities: data-informed planning and improved resource allocation.
- The environment: less avoidable landfill waste, littering, and unnecessary resource consumption.

Teams should support impact claims with data, experiments, prototype demonstrations, or transparent estimates rather than unsupported promises.

8. Official Problem Statement — Shareable Version

Smart and Sustainable Waste Management

Waste management remains a significant environmental and social challenge in campuses and local communities due to improper waste segregation, inefficient collection systems, limited recycling, and inadequate monitoring of waste generation. These challenges lead to environmental pollution, increased waste-disposal costs, and the loss of valuable materials that could otherwise be reused or recycled.

The challenge is to design an innovative, affordable, and scalable technology-driven solution that improves one or more stages of the waste-management process, from waste generation and segregation to collection, recycling, and resource recovery.

Teams may explore approaches such as AI-based waste identification and segregation, smart waste collection and route optimization, recycling and waste-to-resource systems, or data-driven monitoring and prediction of waste generation. Participants are also encouraged to propose original approaches that address the same central challenge.

Expected Outcome: Teams should develop a working prototype, application, AI/ML model, dashboard, simulation, hardware prototype, or other suitable demonstration of their proposed solution. The submission should explain the problem addressed, proposed methodology, practical feasibility, estimated implementation cost, scalability, and potential environmental or social impact.

Solutions will be encouraged to demonstrate how they can reduce waste, improve recycling and resource recovery, optimize collection processes, or support more informed waste-management decisions.

The objective is to move beyond ideas and develop practical solutions that can contribute to cleaner campuses, more sustainable communities, and a circular economy.

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One central challenge. Multiple approaches. Measurable social impact.